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A classification of assembly line balancing problems

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Abstract

Assembly lines are special flow-line production systems which are of great importance in the industrial production of high quantity standardized commodities. Recently, assembly lines even gained importance in low volume production of customized products (mass-customization). Due to high capital requirements when installing or redesigning a line, its configuration planning is of great relevance for practitioners. Accordingly, this attracted attention of many researchers, who tried to support real-world configuration planning by suited optimization models (assembly line balancing problems). In spite of the enormous academic effort in assembly line balancing, there remains a considerable gap between requirements of real configuration problems and the status of research. To ease communication between researchers and practitioners, we provide a classification scheme of assembly line balancing. This is a valuable step in identifying remaining research challenges which might contribute to closing the gap.

Keywords: Configuration of assembly lines; Assembly line balancing; Classification

1. Introduction

An assembly line is a flow-oriented production system where the productive units performing the operations, referred to as stations, are aligned in a serial manner. The workpieces visit stations successively as they are moved along the line usually by some kind of transportation system, e.g., a conveyor belt.

Originally, assembly lines were developed for a cost efficient mass-production of standardized products, designed to exploit a high specialization of labour and the associated learning effects (Shtub and Dar-El, 1989; Scholl, 1999, p. 2). Since the times of Henry Ford and the famous model-T, however, product requirements and thereby the requirements of production systems have changed dramatically. In order to respond to diversified costumer needs, companies have to allow for an individualisation of their products. For example, German car manufacturer BMW offers a catalogue of optional features which, theoretically, results in 10^{32} different models (Meyr, 2004). Multi-purpose machines with automated tool swaps allow for

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facultative production sequences of varying models at negligible setup costs. This makes efficient flow-line systems available for low volume assembly-to-order production (Mather, 1989) and enables modern production strategies like mass-customization (Pine, 1993), which in turn ensures that the thorough planning and implementation of assembly systems will remain of high practical relevance in the foreseeable future.

Due to the high level of automation, assembly systems are associated with considerable investment costs. Therefore, the (re)-configuration of an assembly line is of critical importance for implementing a cost efficient production system. Configuration planning generally comprises all tasks and decisions which are related to equipping and aligning the productive units for a given production process, before the actual assembly can start. This includes setting the system capacity (cycle time, number of stations, station equipment) as well as assigning the work content to productive units (task assignment, sequence of operations).

In light of the high practical relevance, it is not astounding that a massive body of academic literature covers configuration planning of assembly systems. In the scientific discussion, the term assembly line balancing (ALB) is used to subsume optimization models which seek to support this decision process. Since the first mathematical formalization of ALB by Salveson (1955), academic work mainly focused on the core problem of the configuration, which is the assignment of tasks to stations. Because of the numerous simplifying assumptions underlying this basic problem, this field of research was labeled simple assembly line balancing (SALB) in the widely accepted review of Baybars (1986). Subsequent works however, more and more attempted to extend the problem by integrating practice relevant aspects, like u-shaped lines, parallel stations or processing alternatives (Becker and Scholl, 2006). In spite of these efforts, which are referred to as general assembly line balancing (GALB), there seems to be a wide gap between the academic discussion and practical applications. Empirical surveys stemming from the 1970s (Chase, 1974) and 1980s (Schöniger and Spingler, 1989) revealed that only a very small percentage of companies were using a mathematical algorithm for configuration planning at that time. The apparent lack of more recent scientific studies on the application of ALB algorithms indicates that this gap still exists or even has widened.

Three theoretical reasons might explain the aforementioned deficit: (i) Researchers have not considered the "true" real-world problems so far. (ii) The problems were covered, but could not be solved to satisfaction. (iii) Scientific results could not be transferred back to practical applications, e.g., because solutions for special case studies could not be extended to general problems.

Any of these reasons might result from a fundamental problem in communication, which is expressed by an inconsistent use of terms and definitions for the various types of balancing problems (Becker and Scholl, 2006). This is not only impeding the communication within the research community, but also the knowledge transfer to practice.

A first, yet decisive step to resolve this problem lies in a consistent, authoritative classification of ALB problems including all relevant constraints and objectives. A uniform classification enables practitioners to compare their individual problem settings with those covered by research and to single out suitable solution techniques. Furthermore, future research challenges can be identified by structuring the existing body of literature according to the classification scheme. The primary aim of this article is to develop such a classification.

Apparently, the existing distinction of SALB and GALB introduced by Baybars (1986) has become insufficient to reflect the heterogeneity of GALB problems. Especially now, when the problem structure of SALB is well examined and powerful solution techniques exist, it is to be expected that future publications will mainly focus on GALB problems, some of which might show a similar problem structure as SALB, whereas others will deviate considerably. Ghosh and Gagnon (1989) extended Baybars efforts by further distinguishing special characteristics of GALB problems which had been covered by academic work at that time. However, their list lacks a systematic approach and is long outdated because a great variety of additional constraints has been introduced in the meantime. Several researchers have tried to overcome this weakness by developing individual names for their considered problem extensions, which were mostly oriented towards the existing nomenclature. Although these efforts certainly help experts in the field to recognize particular problem characteristics, it might also lead to confusion as long as no guidelines exist as to when an extension is different enough in order to receive a new label or when two extensions with individual names are to be combined. In any case, this policy hardly reveals relations between problems and therefore cannot replace a classification in structuring the literature.

Before a new classification is proposed, the subject to be classified has to be characterized unambiguously. Therefore, Section 2 will start out with a description of the core problem of ALB, which will be used as a basis for the classification scheme presented in Section 3. Then, the classification scheme is applied to structure the existing literature in Section 4. Section 5 provides a first interpretation of achieved results and aims at providing hints on how to close the gap between research and practice by identifying promising areas of future research and characterizing practice relevant problem extensions which have not been covered so far.

2. A definition of assembly line balancing (ALB)

In the following, we define the scope of the classification. First we characterize the SALB problem as the core decision problem in configuration planning in its very basic version. Afterwards, the basic assumptions of SALB are examined of how they have to be adopted for setting the more general assumptions of ALB. That way, a definition of ALB, the field to be classified, can be derived.

Among the family of ALB problems, the best-known and best-studied is certainly the SALB problem. Although it might be far too constrained to reflect the complexity of real-world line balancing, it nevertheless captures its main aspects and is rightfully regarded as the core problem of ALB. In fact, vast varieties of more general problems are direct SALB extensions or at least require the solution of SALB instances in some form. In any case, it is well suited to explain the basic principles of ALB and introduce its relevant terms. A comprehensive review of SALB and its solution procedures is provided by Scholl and Becker (2006).

According to the underlying concept of any SALB formulation, an *assembly line* consists of $k = 1,\dots,m$ (*work*) *stations* arranged along a conveyor belt or a similar mechanical material handling device. The *workpieces* (*jobs*) are consecutively launched down the line and are hence moved on from station to station until they reach the end of the line. A certain set of operations is performed repeatedly on any workpiece which enters a station, whereby the time span between two entries is referred to as *cycle time*. In general, the line balancing problem consists of optimally partitioning (balancing) the assembly work among all stations with respect to some objective. For this purpose, the total amount of work necessary to assemble a workpiece is split up into a set $V = \{1,\dots,n\}$ of elementary operations named *tasks*. Tasks are indivisible units of work and thus each task j is associated with a processing time t_j also referred to as *task time*. Due to technological and/or organizational requirements, tasks cannot be carried out in an arbitrary sequence, but are subject to *precedence constraints*.

The general input parameters of any SALB instance can be conveniently summarized and visualized by a *precedence graph*. This graph contains a node for each task, node weights which equal the task times and arcs reflecting direct as well as paths reflecting indirect precedence constraints. Fig. 1 shows an example precedence graph with $n = 9$ tasks having task times between 2 and 9 (time units).

A feasible *line balance*, i.e., an assignment of tasks to stations, has to ensure that no precedence relationship is violated. The set S_k of tasks assigned to a station k constitutes its *station load* or *work content*, the cumulated task time $t(S_k) = \sum_{j\in S_k} t_j$ is called *station time*.

SALB further assumes that the *cycle time* of all stations is equal to the same value c. Assembly lines with this attribute are called *paced*, as all stations can begin with their operations at the same point in time and also pass on workpieces at the same rate. As a consequence, all station times of a feasible balance may never exceed c, as otherwise the required operations could not be completed before the workpiece leaves the station. Station times can however be smaller than the cycle time, in which case a station k has an unproductive *idle time* of $c - t(S_k)$ time units in each cycle.

For the example of Fig. 1, a feasible line balance with cycle time $c = 11$ and $m = 5$ stations is given by station loads $S_1 = \{1,3\}$, $S_2 = \{2,4\}$, $S_3 = \{5,6\}$, $S_4 = \{7,8\}$ and $S_5 = \{9\}$.

In order to ensure high productivity, any good balance should cause as few idle times as possible.

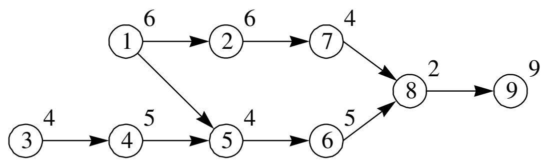

Fig. 1. Precedence graph.

With regard to the objective function considered, SALB problems can further be distinguished (cf. Scholl, 1999, ch. 2.2) into four types: For a given cycle time c, minimizing the sum of station idle times is equal to minimizing the number of opened stations. SALB problems under this objective are called SALBP-1. Conversely, if the number of stations is given, then minimizing the cycle time guarantees minimum idle times, which is known as SALBP-2. If both, number of stations and the cycle time, can be altered, the *line efficiency* E is used to determine the quality of a balance. The line efficiency corresponds to the productive fraction of the line's total operating time t_{sum} and is typically defined as $E = t_{\text{sum}}/(m \cdot c)$. As the total idle time is equal to $t_{\text{sum}} - (m \cdot c)$, a maximization of E also minimizes idle times. The corresponding SALB problem was hence labeled SALBP-E. Finally, the problem of finding a feasible balance for a given number of stations and a given cycle time is known as SALBP-F.

As already mentioned above, all four types of SALB are based on a set of limiting assumptions (c.f. Baybars, 1986; Scholl, 1999, ch. 2.2; Scholl and Becker, 2006):

(S-1) Mass-production of one homogeneous product.
(S-2) All tasks are processed in a predetermined mode (no processing alternatives exist).
(S-3) Paced line with a fixed common cycle time according to a desired output quantity.
(S-4) The line is considered to be serial with no feeder lines or parallel elements.
(S-5) The processing sequence of tasks is subject to precedence restrictions.
(S-6) Deterministic (and integral) task times t_j.
(S-7) No assignment restrictions of tasks besides precedence constraints.
(S-8) A task cannot be split among two or more stations.
(S-9) All stations are equally equipped with respect to machines and workers.

Nearly all of these assumptions have been relaxed or somehow modified by various model extensions considered in the literature. This poses the question which, if any, of the assumptions of SALB is also constitutive for ALB in general.

With regard to real-world line balancing, one of the strongest simplifications of SALB is certainly caused by assumptions (S-2) and (S-9). In real-life, the same task might be performed in different modes and/or with different resources. More powerful machinery might for instance be purchased at higher cost, so that a time-cost trade-off influences the balancing decision. Under the limiting assumptions that processing modes are fixed for any task and each station is equally equipped an explicit consideration of *resources* (machine, men, material and/or tools) is not necessary. As a consequence the decision problem of SALB comes down to assigning tasks to stations. As soon as assumptions (S-2) and (S-9) have to be given up, line balancing needs also to decide on an assignment of required resources to stations along with the respective tasks. At the same time capacity-oriented objectives of SALB become insufficient as resource costs need to be considered.

In order to account for the large variety of ALB problems discussed in the literature, some SALB assumptions need to be modified whereas others are to be given up completely. The following set of assumptions (A-1) to (A-5) is considered to be constitutive for ALB in general (SALB and other problem versions as well are contained as special cases with additional assumptions):

(S-1) → (A-1) The products to be manufactured (one or more) are known with certainty.
(S-2) → (A-2) A set of processing alternatives is given.
(S-3) → (A-3) The line is to be configured such that target production quantities are satisfied for a certain planning horizon. This might be realized by setting the (average) cycle time(s) and, thus, production rate(s) or by seeking to produce as much as possible if maximum sales are not a limiting factor.
(S-4) → (A-4) The line flow is unidirectional.
(S-5) = (A-5) The processing sequence of tasks is subject to precedence restrictions.
(S-6) – (S-9) are given up.

Finally, this leads to the following definition of ALB. All optimization models which aim at supporting that part of assembly line configuration which deals with grouping tasks and their required resources to stations and are in line with the presented five assumptions (A-1)–(A-5) are considered to be assembly line balancing problems. All of those approaches are hence the subject of this work and will be included in the presented classification.

3. The classification scheme for ALB

In the following, we present a scheme for classifying the research field of ALB based on the distinction made in Section 2. Because there are very recent surveys on SALB and GALB research (cf. Scholl and Becker, 2006; Becker and Scholl, 2006) we restrict the presentation to very short descriptions of the relevant characteristics of assembly systems and ALB problems. For further descriptions we refer to, e.g., Buxey et al. (1973), Baybars (1986), Shtub and Dar-El (1989), Ghosh and Gagnon (1989), Erel and Sarin (1998), Scholl (1999, ch. 1), and Rekiek et al. (2002b).

The classification scheme is constructed as follows:

- The assumptions of SALB as outlined in Section 2 are chosen as the common reference for classifying problem characteristics. That means, if not further stated it is supposed that the SALB assumptions apply, so that only deviations from SALB are explicitly provided.
- The basic classification scheme has been adopted from the widely accepted and successful classification scheme for machine scheduling of Graham et al. (1979). Another scheme which successfully helped structuring a complex research field is the one provided by Brucker et al. (1999) for project scheduling, which also employs the tuple-notation.
- As has been established in Section 2, any ALB problem will at least consist of three basic elements: A *precedence graph* which comprises all tasks and resources to be assigned, the *stations* which make up the line and to which those tasks are assigned and some kind of *objective* to be optimized. Accordingly, the presented classification will be based on those three elements which are noted as tuple $[\alpha|\beta|\gamma]$, where:
  α precedence graph characteristics;
  β station and line characteristics;
  γ objectives.

One major advantage of the tuple-notation is that any default value, represented by the symbol $\circ$, can be skipped when a tuple is actually specified. As explained above, the default values of all attributes constitute the SALB problem. In the following notation, the symbol $*$ always indicates that for the respective attribute the alternative values (except for $\circ$) do not exclude each other and can be combined arbitrarily. As all attribute values are chosen such that they are unique, it is not necessary to specify the attribute designators within the tuples.

3.1. Precedence graph characteristics

A precedence graph consists of nodes which represent the tasks of the production process and a set of arcs which represent the precedence relations between the tasks. Moreover, node and arc weights can be considered which reflect important attributes, like task times and processing alternatives, resource requirements and zoning restrictions. These precedence graph characteristics α are represented by the six attributes α_1 to α_6:

3.1.1. Product specific precedence graphs: $\alpha_1 \in \{mix, mult, \circ\}$

This attribute determines whether only a single product and, thus, a single precedence graph is considered or several ones have to be taken into account simultaneously. However, for the line balancing problem it is not the actual number of different products which is decisive, but the degree of homogeneity among precedence graphs.

$\alpha_1 = \text{mix}$. Mixed-model line: Varying models are manufactured on the same production system, the production processes of which are similar enough so that setup times are not present or negligible. Thus, the units of the different models are produced in an arbitrarily intermixed sequence (cf. Scholl, 1999, p. 7). The balancing problem is often based on an average model-mix. In order to anticipate the later *sequencing problem* (cf. Yano and Bolat, 1989) adequately, a *horizontal balancing* objective is usually utilized which attempts to equalize the work content of stations over all models (cf. Scholl, 1999, ch. 3.3 and 3.4; Boysen, 2005, ch. B.2).

$\alpha_1 = \text{mult}$. Multi-model line: Different products are manufactured in batches. Whenever another batch is to be processed, a setup occurs which requires time and resources. Therefore, the ALB problem is not only connected to a (*batch*) *sequencing* but also to a *lot sizing problem* (cf., e.g., Burns and Daganzo, 1987; Dobson and Yano, 1994). However, both additional problems are typically not part of the long/medium-term configuration decisions.

$\alpha_1 = \circ$. A single product is manufactured or the production processes of multiple products are (almost) identical so that they need not be distinguished.

3.1.2. Structure of the precedence graph: $\alpha_2 \in \{spec, \circ\}$

Some research papers restrict themselves to precedence graphs with special structures, mainly for developing more efficient specialized algorithms.

$\alpha_2 = \text{spec}$. The research is restricted to precedence graphs with some kind of special structure, e.g., linear, diverging or converging graphs.

$\alpha_2 = \circ$. The precedence graph can have any acyclic structure.

3.1.3. Processing times: $\alpha_3 \in \{t^{sto}, t^{dy}, \circ\}*$

In reality, task times can vary in time, for instance due to complicated manual operations or defaults of machinery. ALB models can consider this phenomenon in different ways.

$\alpha_3 = t^{\text{sto}}$. Stochastic processing times: It is assumed of processing times follow a known (or even unknown/partially known) distribution function.

$\alpha_3 = t^{\text{dy}}$. Dynamic variations of processing times: These variations are, e.g., due to learning effects of operators.

$\alpha_3 = \circ$. Processing times are considered to be static and deterministic.

3.1.4. Sequence-dependent task time increments: $\alpha_4 \in \{\Delta t_{dir}, \Delta t_{ind}, \circ\}*$

Such task time increments occur whenever the sequence of operations influences their processing times.

$\alpha_4 = \Delta t_{\text{dir}}$. If two tasks are executed at a station one directly after the other, additional time might be required for setup operations or tool changes (Wilhelm, 1999) and repositioning of workpieces (Arcus, 1966; Bautista and Pereira, 2002).

$\alpha_4 = \Delta t_{\text{ind}}$. Indirect sequence-dependent time increments occur if the status achieved by completing particular tasks has an effect on the processing time of other tasks which are executed later in the same or another station (Scholl et al., 2006).

$\alpha_4 = \circ$. Sequence-dependent time increments are not considered.

3.1.5. Assignment restrictions: $\alpha_5 \in \{link, inc, cum, fix, excl, type, min, max, \circ\}*$

Besides precedence relations, assignment restrictions might affect the grouping of tasks to station loads by forcing or forbidding certain combinations.

$\alpha_5 = \text{link}$. Subsets of tasks are linked such that these tasks must be assigned to the same station, e.g., because they employ the same resource which cannot be duplicated.

$\alpha_5 = \text{inc}$. Subsets of tasks are incompatible and must not be assigned to the same station, e.g., because the tasks disturb each other (drilling and measuring) or require the workpiece in incompatible positions.

$\alpha_5 = \text{cum}$. The assignment of tasks to a station is subject to constraints on the cumulated value of particular task attributes (e.g., restricted space for storing material).

$\alpha_5 = \text{fix}$. Some tasks can only be assigned to particular stations, e.g., a required resource cannot be moved when reconfiguring a line.

$\alpha_5 = \text{excl}$. Some tasks must not be assigned to particular stations, e.g., because needed resources cannot be installed there.

$\alpha_5 = \text{type}$. Some tasks have to be assigned to a station from a certain type set, e.g., for working underneath a workpiece.

$\alpha_5 = \text{min}$. When assigning a task, minimum distances to other tasks measured in time, space or sequence positions have to be observed, e.g., for drying the paint.

$\alpha_5 = \text{max}$. Maximum distances between tasks have to be observed, e.g., because a required physical condition can only be maintained for a short time.

$\alpha_5 = \circ$. No assignment restrictions are considered.

3.1.6. Processing alternatives: $\alpha_6 \in \{pa^{\lambda}, \circ\}$

If processing alternatives exist, the production process and, hence the precedence graph is subject to change, so that an additional decision problem arises, which is to select processing alternatives out of the set available (cf. Pinto et al., 1983).

$\alpha_6 = \text{pa}^{\lambda}$. Processing alternatives can be distinguished according to the effects they have on the precedence graph by defining $\lambda \in \{\circ, \text{prec}, \text{subgraph}\}$:

$\lambda = \circ$: Processing alternatives only deviate in processing times and costs (time-cost trade-off).

$\lambda = \mathrm{prec}$: The chosen alternative not only affects times and costs, but also precedence relationships between tasks. For example, an alternative placement of one component at the workpiece might change the precedence relations for installing another.

$\lambda = \mathrm{subgraph}$: Processing alternatives alter complete parts of the production process, so that whole subgraphs are substitutable. This might, e.g., occur whenever a set of options can either be installed separately or completely be replaced by a purchased module.

$\alpha_6 = \circ$. Processing alternatives are not considered.

Remark. Whenever processing alternatives are regarded, the aforementioned precedence graph characteristics (except for α_1 and α_6) might be related to alternatives and tasks rather than tasks only. This can be visualized by attaching $^{\mathrm{pa}}$ to the respective attribute.

3.2. Station and line characteristics

Stations and their arrangement in the assembly line can be classified using the six attributes β_1 to β_6 of the second tuple entry β:

3.2.1. Movement of workpieces: $\beta_1 \in \{\circ\lambda\vartheta, unpac^{\lambda}\}$

$\beta_1 = \circ\lambda\vartheta$. In paced lines, a cycle time restricts the station time at each workstation with $\lambda \in \{\circ, \mathrm{each}, \mathrm{prob}\}$ and $\vartheta \in \{\circ, \mathrm{div}\}$:

$\lambda = \circ$: The (average) work content per cycle of each station over all workpieces is restricted by the cycle time. While this restriction is strict in single-model production, it has just to be fulfilled on average in the case of mixed-model production, some type of parallelization and/or stochastic task times.

$\lambda = \mathrm{each}$: Each single model must strictly fulfill the cycle time restriction in mixed- or multi-model production.

$\lambda = \mathrm{prob}$: The cycle time restriction is obeyed with a given probability (i.e., stochastic task times) or proportion (i.e., uncertain model-mix).

$\vartheta = \circ$: All stations and models have to regard the same global cycle time.

$\vartheta = \mathrm{div}$: (Local) cycle times diverge between stations or models.

$\beta_1 = \mathrm{unpac}^{\lambda}$. An unpaced line is not strictly restricted by a cycle time. Instead, it advances when stations have completed their tasks; with $\lambda \in \{\circ, \mathrm{sync}\}$.

$\lambda = \circ$: Asynchronous workpiece movement: As soon as a station completes its work, the workpiece is moved to the next station or a buffer in front of this station unless blocking occurs. The ALB is accompanied by the additional decision problem of *positioning and dimensioning buffers* (e.g., Baker et al., 1990; Hillier et al., 1993; Malakooti, 1994).

$\lambda = \mathrm{sync}$: In synchronous lines, the movement of workpieces is coordinated between stations. The workpieces are transferred to the respective next station when all stations have completed their current workpiece.

3.2.2. Line layout: $\beta_2 \in \{\circ, u^{\lambda}\}$

$\beta_2 = \circ$. The stations are arranged in a serial manner along the flow of the line.

$\beta_2 = u^{\lambda}$. A U-shaped line layout with crossover stations is used (cf. Miltenburg and Wijngaard, 1994; Monden, 1998; Scholl and Klein, 1999); with $\lambda \in \{\circ, n\}$:

$\lambda = \circ$: The line forms a single U.

$\lambda = n$: An n-U line composed of multiple U-shaped segments is considered (cf. Miltenburg, 1998; Sparling, 1998).

Remark. If other special layouts will be introduced, the classification can easily be extended.

3.2.3. Parallelization: $\beta_3 \in \{pline^{\lambda}, pstat^{\lambda}, ptask^{\lambda}, pwork^{\lambda}, \circ\}*$

$\beta_3 = \mathrm{pline}^{\lambda}$. More than one parallel line is to be balanced or the number of lines installed is part of the decision problem (cf. Geoffrion and Graves, 1976).

$\beta_3 = \mathrm{pstat}^{\lambda}$. When stations are parallelized, their resources and work contents are duplicated so that they process all assigned tasks alternately (cf. Buxey, 1974; Pinto et al., 1981; Sarker and Shanthikumar, 1983; Bard, 1989).

$\beta_3 = \mathrm{ptask}^{\lambda}$. A parallelized task is assigned to more than one station. In addition to their regular work content, stations process the parallelized task interchangeably (cf. Pinto et al., 1975;

Bukchin et al., 2002; Bukchin and Rabinowitch, 2005).

$\beta_3 = \text{pwork}^{\lambda}$. Several working places simultaneously work on the same workpiece at different mounting places such that they do not obstruct each other (Falkenauer, 2005). The balancing problem is connected to a detailed *scheduling problem* which has to avoid that different operators interfere with each other. A special case of a line with parallel workplaces is the *two-sided line* where each station consists of a left-hand-side and a right-hand-side workplace (Bartholdi, 1993; Kim et al., 2000a; Lee et al., 2001).

$\beta_3 = \circ$. Neither type of parallelization is considered.

Remark. If a certain fixed or maximum level of parallelization is constitutive for the problem, this is visualized by the superscript $\lambda \in \{2, 3, \ldots\}$. Such a constraint might be due to technological or space restrictions. For example, the stations of a two-sided line consist of two parallel working places, i.e., $\beta_3 = \text{pwork}^2$. If the level of parallelization is unconstrained or can be set to an arbitrary value by the planner, $\lambda = \circ$ is used.

3.2.4. Resource assignment: $\beta_4 \in \{equip, res^{\lambda}, \circ\}*$

Typically, multiple resources, like operators, machines and tools which provide the skills and/or technological capabilities, forming the station equipment, are necessary to perform tasks at stations.

$\beta_4 = \text{equip}$. For each station exactly one equipment is to be chosen out of a set of prespecified equipment alternatives. So, the balancing problem is connected to an *equipment selection problem* which simultaneously has to decide on the equipment to be installed in each station. In the literature this combined problem is referred to as the *assembly line design problem* (Baybars, 1986).

$\beta_4 = \text{res}^{\lambda}$. Instead of selecting a single equipment out of a set of predetermined equipment alternatives, a stations' equipment is configured along with the task assignment. If several tasks require the same resources, *synergies* can be realized by combining these tasks into the same station loads, because the resources are needed only once and investment costs are at a minimum (cf. Boysen, 2005, pp. 105–108). Two basic types of synergies are denoted with $\lambda \in \{\circ, 01, \max\}*$:

$\lambda = 01$: If more than one task can be performed on the same resource (a tool or machine), investment cost is reduced if those tasks are assigned to the same station, because the resource needs to be installed only once. That is, a 0-1 investment decision has to be made for each station-resource combination (install if any task requires the resource, do not install else).

$\lambda = \max$: If tasks differ with respect to resource quality (concerning, e.g., speed, capabilities, qualification), they require the resources to be selected such that they fulfill the maximum demand level of assigned tasks. For example, if operators require a specific qualification to perform (difficult) tasks, the most challenging task assigned to a station defines the necessary qualification level of the operator and, hence, the wage costs to be paid at that station.

$\lambda = \circ$: Resources are modeled explicitly to account for another type of synergy and/or dependency.

$\beta_4 = \circ$. Resources are not considered explicitly but might influence the balancing decisions via assignment restrictions, processing alternatives, etc. implicitly.

Remark. If resources are modeled explicitly, a part of the task-station assignment restrictions (α_5) might be replaced by *task-resource* and *resource-station* assignment restrictions.

3.2.5. Station-dependent time increments: $\beta_5 \in \{\Delta t_{unp}, \circ\}$

$\beta_5 = \Delta t_{\text{unp}}$. Some part of the station time is consumed by unproductive activities, e.g., workpiece transportation (Bard, 1989), walking distances to change sides in a U-line (Sparling, 1998; Sparling and Miltenburg, 1998) or workers' return to the beginning of a station at the end of the cycle.

$\beta_5 = \circ$. Station-dependent time increments are not regarded.

3.2.6. Additional aspects of line configuration: $\beta_6 \in \{buffer, feeder, mat, change, \circ\}*$

Depending on the production system, additional technical requirements might need to be considered for balancing the line:

$\beta_6 =$ buffer. Buffer storages are required and have to be allocated and dimensioned.

$\beta_6 =$ feeder. One or more feeder lines which flow into a main line necessitate a simultaneous coordination of task assignments and cycle time (Hautsch et al., 1972; Lapierre and Ruiz, 2004).

$\beta_6 =$ mat. Boxes which contain the required material need to be positioned and dimensioned (cf. Bukchin and Meller, 2005).

$\beta_6 =$ change. If certain tasks require the workpiece to be in a particular position (lifted, tilted, etc.), a decision has to be made, whether positions are fixed within stations or special machines are assigned to stations which allow for a change of positions.

$\beta_6 = \circ$. No additional aspects of line configuration are regarded.

3.3. Objectives

Finally, the optimization of ALB will be guided by some objective which evaluates solutions. In the case of multi-objective optimization more than a single objective can be selected out of the set $\gamma \in \{m, c, E, \mathrm{Co}, \mathrm{Pr}, \mathrm{SSL}^{\lambda}, \mathrm{score}, \circ\}*$.

$\gamma = m$. Minimize the number of stations m subject to a given output target for a certain planning horizon (specified by the cycle time c or the production rate). In case that parallel stations or workplaces are included in m, the number of sequential stages might be restricted (maximal line length).

$\gamma = c$. Minimize the cycle time c or, equivalently, maximize the production rate subject to a given number of stations m.

$\gamma = E$. Maximize the line efficiency E (productive part of the line capacity) such that restrictions on the production rate and/or the number of stations are fulfilled.

$\gamma = \mathrm{Co}$. Cost minimization for a given output target. A classification of cost factors shall not be provided as cost types can be assigned to various elements of the assembly system, like tasks, stations, processing alternatives and/or resources, depending on the considered model formulation. Wage costs might, e.g., depend on the selected processing alternative, but can also be directly assigned to a task as long as no alternatives exist. If each station is equipped with a worker who can perform any task, wage costs might be directly assigned to stations. An in-depth discussion of cost factors in ALB is provided by Steffen (1977) and Amen (1997), Amen (2000a) Amen (2000b), Amen (2001), Amen (2006).

$\gamma = \mathrm{Pr}$. The profit, which is defined as the difference between the revenue (which depends on the production rate and, hence, the cycle time) and the costs, is maximized. A further distinction shall not be provided because of the aforementioned issue of cost type assignment.

$\gamma = \mathrm{SSL}^{\lambda}$. Instead or in addition to the objectives mentioned above, the station times should be smoothed. Two ways of smoothing are given by $\lambda \in \{\mathrm{stat}, \mathrm{line}\}$:

$\lambda =$ stat: In mixed-model production the varying station times caused by different models are to be smoothed (horizontal balancing; Merengo et al., 1999).

$\lambda =$ line: Station times are balanced over all stations of the line (vertical balancing; Rachamadugu and Talbot, 1991).

$\gamma =$ score. The objective is to minimize or maximize some composite score which is related to one or more attributes describing bottleneck aspects or further measures of efficiency, e.g., required grip strengths, quality or number of workpiece position changes.

$\gamma = \circ$. No objective function is required, only feasible solutions are searched for.

4. Classifying the literature

With this scheme on hand, we classify that part of the ALB literature which deals with GALB problems. For the versions of SALB, we only specify the tuple-notation, because there is no need to structure this field of research (instead, we refer to Scholl and Becker, 2006):

SALBP-F: $[\,|\,|\,]$ SALBP-1: $[\,|\,|m]$
SALBP-2: $[\,|\,|c]$ SALBP-E: $[\,|\,|E]$

The following table assigns the tuple-notation to each relevant publication and its contribution to the research field of GALB. If more than one problem

M	mathematical model	B	bound computation
SA	sensitivity/ stability analysis	HS	heuristic start procedure for initial solution, mostly priority rule based
HI	heuristic improvement procedure		
Exact solution procedures:		Meta-heuristics:	
DP	dynamic programming	TS	tabu search
B&B	branch & bound	GA	genetic algorithm
GR	graph theoretic approach	SA	simulated annealing
B&C	branch & cut	ANT	ant system approach
CG	column generation	GRASP	greedy randomized adaptive search procedure
ENU	enumeration approach		

version is treated in a paper the most complex tuple-notation is reported. Further we distinguish between the following contributions made by the respective publication:

Remark. The following list does not include the numerous papers which treat related decision problems of configuration planning which assume a *given* line balance. For example, a massive body of literature covers the buffer allocation problem in unpaced asynchronous lines (see the review papers of Dallery and Gershwin, 1992; Papadopoulos and Heavey, 1996; Gershwin, 2000). Since a change in station loads typically affects optimal buffer allocation considerably, a simultaneous optimization would be desirable to identify the systems' global optimum. However, such simultaneous approaches are currently not on-hand.

Publication	Notation	Contribution
Aase et al. (2003), Aase et al. (2004)	$[\|u\|m]$	M, B, B&B
Agnetis et al. (1995)	$[\text{spec, inc, fix}\|\|\text{SSL}^{\text{line}}]$	DP
Agrawal (1985)	$[\|\|\text{score}]$	HS
Ajenblit and Wainwright (1998)	$[\|u\|m, \text{SSL}^{\text{line}}]$	GA
Akagi et al. (1983)	$[\|\text{pwork}\|]$	HS
Amen (1997), Amen (2000a), Amen (2000b), Amen (2001), Amen (2006)	$[\|\text{res}^{\max}\|\text{Co}]$	M, B, B&B, HS
Arcus (1966)	$[\text{mix}, \Delta t_{\text{dir}}, \text{cum, fix}\|\text{res}^{\max}, \Delta \text{t}_{\text{unp}}, \text{pwork}\|\text{E}]$	HS
Askin and Zhou (1997)	$[\text{mix}\|\text{pstat}\|\text{Co}]$	M, B, HS
Bard (1989)	$[\|\text{pstat, ptask}, \Delta \text{t}_{\text{unp}}\|\text{Co}]$	DP
Bartholdi (1993)	$[\text{fix, type}\|\text{pwork}^2\|m]$	HS
Bautista and Pereira (2002)	$[\text{inc}, \Delta t_{\text{dir}}\|\|m, \text{score}]$	ANT
Bautista and Pereira (2006)	$[\text{cum}\|\|m, c, \text{score}]$	M, HI, ANT
Bautista et al. (2000)	$[\text{inc}\|\|m, \text{score}]$	HS, GRASP
Baykasoglu and Özbakir (2006)	$[t^{\text{sto}}\|\text{prob}, u\|m]$	GA
Berger et al. (1992)	$[\text{spec}\|u\|m]$	B, B&B
Bhattacharjee and Sahu (1987)	$[\text{link, inc, fix}\|\text{station}\|m, \text{SSL}^{\text{line}}]$	HS
Boysen and Fliedner (2006)	$[t^{\text{sto}}, \text{link, inc, cum, pa}\|u, \text{pstat, ptask, res}^{01}, \text{res}^{\max}\|\text{Pr}]$	HS, GR
Bukchin and Rabinowitch (2005)	$[\text{mix}\|\text{div, ptask, res}^{01}\|\text{Co}]$	M, B, B&B
Bukchin and Rubinovitz (2003)	$[\text{pa}\|\text{pstat, equip}\|\text{Co}]$	M, B, B&B
Bukchin and Tzur (2000)	$[\text{pa}\|\text{equip}\|\text{Co}]$	M, B, B&B

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Table (*continued*)

Publication	Notation	Contribution
Bukchin et al. (1997), Bukchin and Masin (2004)	[$\mid$pwork$\mid m$, score]	B, B&B
Bukchin et al. (2002)	[mix$\mid$ptask$\mid$score]	M, HS, HI
Buxey (1974)	[Δt_{dir}, link, inc, max$\mid$pstat$\mid$score]	HS
Capacho and Pastor (2004)	[pa$^{\text{subgraph}}\mid\mid m$]	M
Carnahan et al. (2001)	[cum$\mid\mid c$, score]	HS, GA
Carraway (1989)	[$t^{\text{sto}}\mid$prob$\mid m$]	DP
Carter and Silverman (1984)	[$t^{\text{sto}}\mid\mid$Co]	HS
Chakravarty (1988)	[$t^{\text{dy}}\mid\mid E$]	HS, DP
Chakravarty and Shtub (1985)	[mult$\mid$div$\mid$Co]	M, HS
Chakravarty and Shtub (1986)	[mult, $t^{\text{sto}}\mid$div$\mid$Co]	HS
Chiang and Urban (2002)	[$t^{\text{sto}}\mid$prob$\mid m$]	HS, HI
Dar-El and Rabinovitch (1988)	[mult, $t^{\text{dy}}\mid\mid$Co]	M
Deckro (1989)	[link, inc, max$\mid\mid m$, c]	M
Dolgui and Ihnatsenka (2004), Dolgui et al. (1999), Dolgui et al. (2001a), Dolgui et al. (2001b) Dolgui et al. (2001c), Dolgui et al. (2003), Dolgui et al. (2006)	[link, inc$\mid$pwork$\mid$Co]	M, HS, *B*, GR, B&B
Domschke et al. (1996)	[$\mid\mid E$, SSL$^{\text{stat}}$]	B&B
Erel and Gökcen (1999), Gökcen and Erel (1998)	[mix$\mid$div$\mid m$]	M, GR
Erel et al. (2001)	[$\mid u\mid m$]	SA
Erel et al. (2005)	[$t^{\text{sto}}\mid u\mid$Co]	HS
Gadidov and Wilhelm (2000)	[type, pa$\mid$res$^{01}\mid$Co]	M, HS, B&C
Gamberini et al. (2004)	[$t^{\text{sto}}\mid u\mid$Co, score]	HS
Gökcen and Agpak (2006)	[$\mid u\mid m$, c, score]	M
Gökcen and Erel (1997)	[mix, link, inc$\mid\mid$score]	M
Gökcen et al. (2005)	[$\mid u\mid m$]	M, GR
Haq et al. (2005)	[mix$\mid\mid m$]	HS, GA
Hautsch et al. (1972)	[link$\mid$pwork^2, res$^{\text{max}}$, feeder$\mid E$]	HS
Henig (1986)	[$t^{\text{sto}}\mid$prob$\mid$Co]	DP
Johnson (1983)	[type$\mid$div, ptask$\mid m$]	B, B&B
Johnson (1991)	[type$\mid\mid m$]	B, B&B
Kao (1976), Kao (1979)	[$t^{\text{sto}}\mid$prob$\mid m$]	DP
Karabati and Sayin (2003)	[mix$\mid$unpac$^{\text{sync}}\mid$score]	M, HS
Karini and Herer (1995)	[mult, $t^{\text{dy}}\mid$unpac$^{\text{sync}}\mid$score]	M, HS
Kim and Park (1995)	[cum$\mid$equip$\mid m$]	M, HS, B&C
Kim et al. (2000a)	[fix, type$\mid$pwork$^2\mid m$]	GA
Kim et al. (2000b), Kim et al. (2000c), Kim et al. (2006)	[mix$\mid u\mid$SSL$^{\text{line}}$]	GA
Kimms (2000)	[mult, spec$\mid$unpac, equip$\mid$Co]	M, B, CG
Klenke (1977)	[t^{sto}, link, inc$\mid$prob$\mid$Pr]	M, ENU
Kottas and Lau (1973), Kottas and Lau (1976), Kottas and Lau (1981)	[$t^{\text{sto}}\mid\mid$Co]	HS
Lapierre and Ruiz (2004)	[link, inc, type$\mid$div, pwork^2, feeder$\mid m$]	HS
Lapierre et al. (2006)	[type$\mid$pwork$^2\mid m$]	TS
Lee et al. (2001)	[type$\mid$pwork$^2\mid$score]	HS
Leu et al. (1994)	[fix$\mid\mid$score]	HS, GA
Levitin et al. (2006)	[pa$\mid$equip$\mid c$]	GA

Table (*continued*)

Publication	Notation	Contribution
Liu and Chen (2002)	[cum | | c]	M, HS
Lyu (1997)	[t^{sto} | | Co]	HS
Macaskill (1972)	[mix | | E]	HS
Malakooti (1991)	[| | m, c, Co]	M, ENU
Malakooti (1994)	[| unpac, buffer | m, c, Co, score]	HS
Malakooti and Kumar (1996)	[fix | unpac, buffer | m, c, Co, score]	HS
Matanachai and Yano (2001)	[mix | | SSL^{line}, SSL^{stat}]	HS, HI
McMullen and Frazier (1997), McMullen and Frazier (1998), McMullen and Tarasewich (2003), McMullen and Tarasewich (2006)	[mix, t^{sto} | pstat | Co, SSL^{line}, score]	SA, ANT
Merengo et al. (1999)	[mix | each | m, SSL^{line}, SSL^{stat}]	HS, HI
Miltenburg (1998)	[fix | u^n, Δt_{unp} | m, score]	DP
Miltenburg (2002)	[mix | u | SSL^{line}, SL^{stat}]	GA
Miltenburg and Wijngaard (1994)	[| u | m]	HS, DP
Miralles (2005)	[pa, link, cum | equip | c]	M, B, B&B
Moodie and Young (1965)	[t^{sto} | | SSL^{line}]	HS
Nicosia et al. (2002)	[pa | equip | Co]	B, DP
Nkasu and Leung (1995)	[t^{sto} | prob | m, c, E]	M
Park et al. (1997)	[spec, inc, pa^{prec} | | c]	HS, HI
Pastor and Corominas (2000)	[link, inc, type, max | | SSL^{line}]	M, HS, TS
Pastor et al. (2002)	[mult, cum, fix | | c, SSL^{line}, SSL^{stat}]	HS, TS
Pinnoi and Wilhelm (1997a)	[mult, link, inc, cum, type, pa | div, equip, pstat, pwork | Co]	M
Pinnoi and Wilhelm (1997b)	[| | SSL^{line}]	M, B, BC
Pinnoi and Wilhelm (1998)	[pa | equip | Co]	M, B, BC, HS
Pinto et al. (1975)	[| ptask | Co]	M, B, B&B
Pinto et al. (1981)	[| pstat | Co]	M, B, B&B
Pinto et al. (1983)	[pa | | Co]	M, B, B&B
Ponnambalam et al. (2000)	[| | m, E, SSL^{line}]	GA
Rachamadugu and Talbot (1991)	[| | SSL^{line}]	M, HI
Raouf and Tsui (1982)	[t^{sto}, fix, excl | prob | m, SSL^{stat}]	HS
Reeve and Thomas (1973)	[t^{sto} | | SSL^{line}]	HS
Rekiek et al. (2001)	[Δt_{dir}, link, inc, fix, type, pa | | Co, SSL^{line}]	GA
Rekiek et al. (2002a)	[link, fix, pa | | Co, SSL^{line}]	GA
Roberts and Villa (1970)	[mix, link | each, ptask | m]	M, GR
Rosenberg and Ziegler (1992)	[| res^{max} | Co]	M, B, HS
Rosenblatt and Carlson (1985)	[| | Pr]	M, HS, ENU
Rubinovitz and Bukchin (1993)	[pa | equip | m]	B, B&B
Sabuncuoglu et al. (2000)	[| | m, SSL^{line}]	GA
Sarin and Erel (1990)	[t^{sto} | | Co]	DP
Sarin et al. (1999)	[t^{sto} | | Co]	HS, HI
Sarker and Shanthikumar (1983)	[| pstat | Co, SSL^{line}]	HS, HI
Sawik (2002)	[mix, cum | div, ptask | c]	M
Schofield (1979)	[mix, link, inc, fix, excl | Δt_{unp} | c]	HS
Scholl and Becker (2005)	[| res^{max} | Co]	M, B, B&B
Scholl and Klein (1999)	[| u | E]	B, B&B
Scholl et al. (2006)	[Δt_{ind} | | m]	M, ENU

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Table (*continued*)

Publication	Notation	Contribution
Shin and Min (1991)	$[t^{\text{sto}}\,\vert\,\text{prob}\,\vert\,\text{Co}]$	HS
Shtub (1984)	$[t^{\text{sto}}, \text{pa}\,\vert\,\vert\,\text{Co}]$	M, HS
Shtub and Dar-El (1990)	$[\,\vert\,\vert\,m, c, \text{score}]$	M, HS
Silverman and Carter (1986)	$[t^{\text{sto}}\,\vert\,\vert\,\text{Co}]$	HS
Sniedovich (1981)	$[t^{\text{sto}}\,\vert\,\text{prob}\,\vert\,m]$	DP
Sotskov et al. (2006)	$[t^{\text{sto}}\,\vert\,\vert\,m]$	SE
Sparling (1998)	$[\,\vert\,u^n, \text{pline}, \Delta t_{\text{unp}}\,\vert\,m]$	M, HS
Sparling and Miltenburg (1998)	$[\text{mix}\,\vert\,u, \Delta t_{\text{unp}}\,\vert\,m, \text{SSL}^{\text{stat}}]$	M, HS
Sphicas and Silverman (1976)	$[t^{\text{sto}}\,\vert\,\text{prob}\,\vert\,m]$	M
Süer (1998)	$[\,\vert\,\text{pline}, \text{pstat}\,\vert\,m, c, \text{score}]$	HS
Suresh and Sahu (1994), Suresh et al. (1996)	$[t^{\text{sto}}\,\vert\,\vert\,\text{SSL}^{\text{line}}, \text{score}]$	SA, GA
Thomopoulos (1970)	$[\text{mix}\,\vert\,\vert\,m, \text{SSL}^{\text{stat}}]$	HS
Tsujimura et al. (1995)	$[t^{\text{sto}}\,\vert\,\vert\,\text{score}]$	M, GA
Ugurdag et al. (1997)	$[\,\vert\,\vert\,c, \text{SSL}^{\text{line}}]$	M, HS, HI
Urban (1998)	$[\,\vert\,u\,\vert\,m]$	M
Urban and Chiang (2006a)	$[t^{\text{sto}}\,\vert\,\text{prob}, u, \Delta t_{\text{unp}}\,\vert\,m]$	M, B
Urban and Chiang (2006b)	$[t^{\text{sto}}\,\vert\,\text{unpac}^{\text{async}}\,\vert\,m]$	HS, TS
Van Hop (2006)	$[\text{mix}, t^{\text{sto}}\,\vert\,\text{div}\,\vert\,m]$	M, HS, HI
Vilarinho and Simaria (2002)	$[\text{mix}, \text{link}, \text{inc}\,\vert\,\text{prob}, \text{pstat}\,\vert\,m, \text{SSL}^{\text{line}}, \text{SSL}^{\text{stat}}]$	M, SA
Vilarinho and Simaria (2006)	$[\text{mix}, \text{link}, \text{inc}, \text{fix}\,\vert\,\text{pstat}\,\vert\,m, \text{SSL}^{\text{line}}, \text{SSL}^{\text{stat}}]$	M, ANT
Visich et al. (2002)	$[\text{mix}\,\vert\,u\,\vert\,\text{SSL}^{\text{stat}}]$	TS, SA
Wilhelm (1999)	$[\Delta t_{\text{dir}}, \text{pa}\,\vert\,\text{equip}\,\vert\,\text{Co}]$	M, *B*, CG
Wilhelm and Gadidov (2004)	$[\text{mix}, \text{cum}, \text{pa}\,\vert\,\text{res}^{01}\,\vert\,\text{Co}]$	M, HS, BC
Wilson (1986)	$[t^{\text{sto}}, \text{pa}\,\vert\,\vert\,\text{Co}]$	M
Zäpfel (1975)	$[\text{link}, \text{inc}\,\vert\,\vert\,\text{Pr}]$	M, ENU

5. Research challenges and conclusions

The classification of the literature reveals a number of important open fields of research which require an in-depth discussion to narrow the gap between research and practice. In the following, we discuss those which seem to be most important.

5.1. Rebalancing

Within the lifetime of a modern production system, ALB problems do not only occur once prior to its construction, but rather continuously as *reconfiguration* or *rebalancing* (Schofield, 1979; Falkenauer, 2005). These might be necessary to react on shifts in the demand structure or whenever new production technologies become available. As a consequence, stations will often have a unique identity expressed by a position in the workshop, assigned resources or space restrictions ($\alpha_5 = \text{cum}$), instead of being abstract entities. In the literature, reconfiguration is often considered by introducing assignment restrictions. If machinery is too heavy for being moved to another station, the assignment of tasks which require such a machine can be forced to its current station by an assignment restriction ($\alpha_5 = \text{fix}$). Nevertheless, often the reallocation of machines is not technically impossible, but rather associated with moving costs (Gamberini et al., 2004). The same applies to operators at a station, who might need to be trained with regard to the new work content. Costs can be decreased when workers keep as much of their previous tasks as possible. Accordingly, assembly line balancing should regard these *moving costs*.

5.2. Cost synergies

A similar phenomenon occurs when several tasks require the same resources. In this case, cost of capital can be decreased when tasks are assigned to the same station, which is typically enforced by zoning restrictions ($\alpha_5 = \text{link}$). Whether the benefits associated with avoiding an extra resource actually surmount the disadvantages of a more restricted balancing problem, can just be answered by explicitly modeling *cost synergies* ($\beta_4 = \text{res}^{01}$).

In the literature, these cost aspects are often neglected and costs are directly assigned to tasks or stations instead. This, however, does often not reflect the real situation. In real life, costs only arise by the purchase or usage of a resource which is positioned at a station. By exactly assigning costs to the resources instead of stations or tasks, a more realistic modeling of the decision problem could be attained.

5.3. Sequence-dependent time increments

Sequence-dependent relations between tasks also require further investigation. Occasionally, precedence-dependent time increments (Buxey, 1974; Wilhelm, 1999; Rekiek et al., 2001; Bautista and Pereira, 2002) are regarded, which result from a tool or position change between predecessors and their direct successors ($\alpha_4 = \Delta t_{\text{dir}}$). However, task times can also be influenced by an indirect succession of tasks ($\alpha_4 = \Delta t_{\text{ind}}$). These time increments occur if the resulting transformation of a completed task obstructs the implementation of another, e.g., seat and safety belt of a car. These dependencies can even cover more than two tasks. For example, the assembly of a car's hand-break might be obstructed only after both front seats have been installed. Otherwise, the access is free from at least one side. In an extreme case, the assembly of a hand break might even be impossible if both seats are installed. These phenomena cannot be included in traditional arcs of the precedence graph, since they depict dependencies exclusively between two tasks. By disregarding these dependencies assembly line balancing might result in infeasible line balances.

5.4. Feeder lines

Little attention has been paid to the so called "additional aspects of line configuration". Real world assembly lines often have one or more *feeder lines* ($\beta_5 = \text{feeder}$) which flow into a main line (Tempelmeier, 2003). A possible approach to configure such an assembly system is to balance the main line first and use the retrieved cycle time to balance each feeder line separately. Whether this decomposition results to a global optimum of the whole assembly system is questionable, especially if operators can perform tasks on both the feeder and the main line at their contact point (Lapierre and Ruiz, 2004).

5.5. Material supply

Another important but almost completely ignored aspect is the *material supply* of an assembly line ($\beta_5 = \text{mat}$). Material is usually provided in some kind of container along the line. A container could possibly be some traditional shelf or box, but also an automated guided vehicle (AGV) which delivers the material directly to the stations. In such a setting, new decision problems arise concerning the dimensions of these containers for different components (Bukchin and Meller, 2005) and their allocation along the line. If more than one station is served by the same material container or if distances vary depending on the actual position of the operator, station times are affected by variable walking distances for fetching the material. Due to these interdependencies, it might be necessary in practice to solve line balancing and material supply problems simultaneously.

5.6. Parallel working places

Furthermore, little attention has been paid to *parallel working places* ($\beta_3 = \text{pwork}$). Although, simultaneous work at a station requires a sufficiently large workpiece, there are important practical applications in automobile and related industries (Falkenauer, 2005). Workers are able to perform operations at different mounting places at the same time if tasks are compatible and do not require the workpiece to be in an exclusive position. In case of parallel work, precedence constraints lead to additional idle times whenever one operator has to wait for another one finishing a preceding task. Accordingly, the station time can no longer be computed by simply summing up the assigned task times, but requires the solution of a detailed scheduling problem instead. To close the gap to practice in this field of research, optimization models have to be developed which go far beyond a simple assignment of tasks to working places.

5.7. Processing alternatives

Over the last few years, a lot of research has regarded *processing alternatives*. Most tasks can be performed in different ways: A faster machine and/or multiple manning decrease the processing time but commonly raise cost of capital (time-cost trade-off). Choosing processing alternatives and balancing the line simultaneously promises better configurations than deciding successively. A successive planning approach selects a processing alternative for each task first and then balances the line with regard to the chosen alternatives. Hints on the advantage's extent when planning simultaneously (maybe with less efficient algorithms) in comparison to planning successively would be of great value in this field of research.

Two different approaches have been proposed to incorporate processing alternatives into ALB. The first is known as the *equipment selection problem* [pa | equip | Co] (e.g., Pinnoi and Wilhelm, 1998; Bukchin and Tzur, 2000). It is based on the assumption that there is a fixed set of equipments (complete set of resources) exactly one of which has to be selected and assigned to a station. The alternative approach consists in *assigning processes to tasks*. In addition to line balancing, for each task exactly one processing alternative has to be chosen out of a set of possible ones. Because these processes require resources, the problem can be interpreted as an (implicit) *equipment design problem*, represented by [pa | | Co] in our classification scheme (e.g., Pinto et al., 1983). The chosen processes are usually not independent of each other, which has to be reflected by considering possible synergies arising from jointly using resources for several tasks at a station ($\beta_4 = \text{res}^{01}$) or different types of assignment restrictions ($\alpha_5 = \text{link}^{\text{pa}}$, $\alpha_5 = \text{inc}^{\text{pa}}$, $\alpha_5 = \text{cum}^{\text{pa}}$).

Both concepts can in principle be used interchangeably. However, the resulting mathematical models might differ considerably. While the equipment selection approach excludes incompatible processing alternatives prior to modeling at the cost of a potentially large number of equipment alternatives, the equipment design approach needs to exclude incompatible combinations within the model which can hence lead to more restrictions and a higher complexity. Due to limitations in data generation and model solving, both approaches have to be limited to a subset of possible alternatives in real-world settings. In order to find out which approach is better suited to different situations, comparative analyses should be performed.

In the literature, both approaches are so far restricted to covering different modes of accomplishing a task. In reality, processing alternatives might, however, also alter precedence constraints between tasks ($\alpha_6 = \text{pa}^{\text{prec}}$). For example, the automated assembly of an option using a robot can be obstructed by an already installed part whereas a manual assembly is still possible. Even whole subgraphs of the precedence graph may be substituted by a processing alternative ($\alpha_6 = \text{pa}^{\text{subgraph}}$). An example is the assembly of a procured module which unites a set of optional features instead of individually installing each option. Both approaches of processing alternatives need to be extended in this direction.

5.8. Disassembly line balancing

A related problem to ALB labeled disassembly line balancing (DLB) raised in the recent years within the field of reverse logistics (Fleischmann et al., 1997). Though a lot of classified ALB characteristics are relevant for DLB, too, some important peculiarities exist, which are exclusively related to the DLB-case. In contrast to ALB, where a finished product leaves the assembly system each production cycle, a returned product is disassembled into its consisting parts, which build the output of the disassembly process. Moreover, hazardous tasks may damage demanded parts disassembled later on in the process, which leads to sequence-dependent output rates. Excellent overviews about the peculiarities of DLB are given by Güngör and Gupta (2002) as well as Lambert and Gupta (2005). To avoid continual changes of the viewpoint between the ALB and DLB perspective we consciously exclude DLB from our classification within this paper. Nevertheless, an enhancement of our classification scheme to the specific features of DLB would be a valuable contribution of further research.

5.9. Test beds and flexible solution procedures

Our survey finally revealed that there is a dire need of systematic evaluations to identify those decision concepts – combining a (theoretical) problem type considered, the mathematical model derived and the (exact or heuristic) solution method applied – which are best suited to solving real-world assembly line balancing problems. In particular, this includes the systematic generation of realistic test beds and the development of solution algorithms flexible enough to jointly cover as many problem characteristics as possible. The structure provided in this paper might therefore also be employed to develop new instance generators and solution procedures which cover all or a certain systematic subset of important problem characteristics.

We would thus like to encourage all researchers to make use of the provided scheme (and to extend it if necessary), in order to establish a common reference for past as well as future publications concerning assembly line balancing. This should ease the coordination of remaining steps to close the stated gap between research and practice substantially.

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